

Started updating as on July 18th, 2024 with Prev. Versions

Create an Angular 20 application that shows employee records in a table with the following features:

- Display employee records in a table
- Edit employee record (inline or with form)
- Delete employee record
- Add new employee using a form

using a **service + REST API** (instead of local array).

Application Flow

1. **EmployeeService** → Handles API calls (CRUD) using HttpClient.
2. **EmployeeComponent** → UI (table + form). Calls service methods.
3. **Backend** → For testing, you can use json-server as REST API.

Create a basic Angular 20 Application

```
>ng new empMgtApp
```

Open it in Visual Studio Code and invoke Terminal.

Install Bootstrap

```
> npm install bootstrap
```

```
D:\ANG-2025\Day10-EmpCrud\empMgtApp>npm i bootstrap
added 2 packages, and audited 704 packages in 2s

123 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

D:\ANG-2025\Day10-EmpCrud\empMgtApp>
```

In angular.json → add bootstrap CSS:

```
"styles": [
  "node_modules/bootstrap/dist/css/bootstrap.min.css",
  "src/styles.css"
]
```

Mock Backend (JSON Server): Install JSON server (if not already):

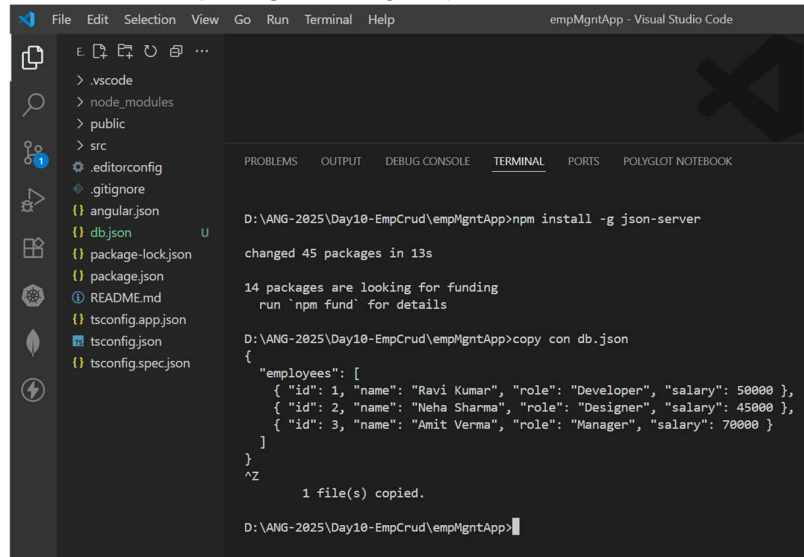
```
>npm install -g json-server
```

Create db.json:

I am using here in the terminal a copy con command to create file on command prompt:

```
{
  "employees": [
    { "id": 1, "name": "Ravi Kumar", "role": "Developer", "salary": 50000 },
    { "id": 2, "name": "Neha Sharma", "role": "Designer", "salary": 45000 },
    { "id": 3, "name": "Amit Verma", "role": "Manager", "salary": 70000 }
  ]
}
```

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```
File Edit Selection View Go Run Terminal Help empMgntApp - Visual Studio Code

> .vscode
> node_modules
> public
> src
  editorconfig
  gitignore
  angular.json
  db.json
  package-lock.json
  package.json
  README.md
  tsconfig.app.json
  tsconfig.json
  tsconfig.spec.json

D:\ANG-2025\Day10-EmpCrud\empMgntApp>npm install -g json-server

changed 45 packages in 13s

14 packages are looking for funding
  run `npm fund` for details

D:\ANG-2025\Day10-EmpCrud\empMgntApp>copy con db.json

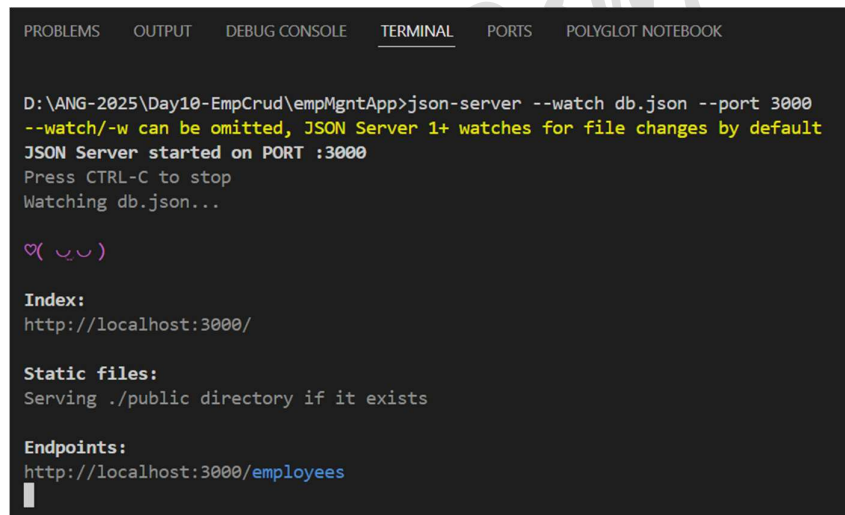
{
  "employees": [
    { "id": 1, "name": "Ravi Kumar", "role": "Developer", "salary": 50000 },
    { "id": 2, "name": "Neha Sharma", "role": "Designer", "salary": 45000 },
    { "id": 3, "name": "Amit Verma", "role": "Manager", "salary": 70000 }
  ]
}
^Z

1 file(s) copied.

D:\ANG-2025\Day10-EmpCrud\empMgntApp>
```

Run server:

`json-server --watch db.json --port 3000`



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POLYGLOT NOTEBOOK

D:\ANG-2025\Day10-EmpCrud\empMgntApp>json-server --watch db.json --port 3000
--watch/-w can be omitted, JSON Server 1+ watches for file changes by default
JSON Server started on PORT :3000
Press CTRL-C to stop
Watching db.json...

♡( ͡° ͜ʖ ͡° )

Index:
http://localhost:3000/

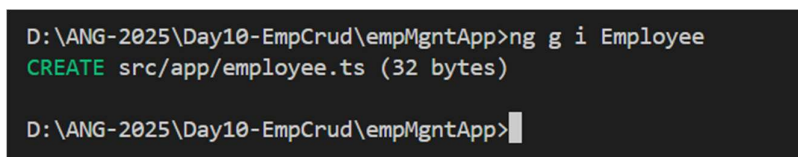
Static files:
Serving ./public directory if it exists

Endpoints:
http://localhost:3000/employees
```

REST API Endpoints:

- GET `http://localhost:3000/employees` → All employees
- GET `http://localhost:3000/employees/:id` → Employee by ID
- POST `http://localhost:3000/employees` → Add employee
- PUT `http://localhost:3000/employees/:id` → Update employee
- DELETE `http://localhost:3000/employees/:id` → Delete employee

To manage Employee object lets create an interface named Employee



```
D:\ANG-2025\Day10-EmpCrud\empMgntApp>ng g i Employee
CREATE src/app/employee.ts (32 bytes)

D:\ANG-2025\Day10-EmpCrud\empMgntApp>
```

Create Employee Interface

```
export interface Employee {  
  id?: number;  
  name: string;  
  role: string;  
  salary: number;  
}
```

Create EmployeeService

```
D:\ANG-2025\Day10-EmpCrud\empMgtApp>ng g s services/EmployeeService  
  
Would you like to share pseudonymous usage data about this project with the Angular Team  
at Google under Google's Privacy Policy at https://policies.google.com/privacy. For more  
details and how to change this setting, see https://angular.dev/cli/analytics.  
  
No  
Global setting: enabled  
Local setting: disabled  
Effective status: disabled  
CREATE src/app/services/employee-service.spec.ts (383 bytes)  
CREATE src/app/services/employee-service.ts (127 bytes)
```

Code for EmployeeService

```
import { Injectable } from '@angular/core';  
import { HttpClient } from '@angular/common/http';  
import { Observable } from 'rxjs';  
import { Employee } from '../employee';  
  
@Injectable({  
  providedIn: 'root'  
})  
export class EmployeeService {  
  private apiUrl = 'http://localhost:3000/employees';  
  
  constructor(private http: HttpClient) { }  
  
  // Get all employees  
  getEmployees(): Observable<Employee[]> {  
    return this.http.get<Employee[]>(this.apiUrl);  
  }  
  
  // Get employee by id  
  getEmployeeById(id: any): Observable<Employee> {  
    return this.http.get<Employee>(`${this.apiUrl}/${id}`);  
  }  
}
```

```
// Add employee
addEmployee(emp: Employee): Observable<Employee> {
  return this.http.post<Employee>(this.apiUrl, emp);
}

// Update employee
updateEmployee(id: any, emp: Employee): Observable<Employee> {
  return this.http.put<Employee>(`${this.apiUrl}/${id}`, emp);
}

// Delete employee
deleteEmployee(id: any): Observable<void> {
  return this.http.delete<void>(`${this.apiUrl}/${id}`);
}
}
```

We'll create following components:

- **Employee List** → Show all employees with actions (edit, delete, view details).
- **Add Employee** → Form to add a new employee.
- **Edit Employee** → Form to edit an existing employee.
- **Employee Details** → Show employee details by ID.

And we'll connect them with **Angular Router**.

Let's create components so routing works fine:

>ng g c components/EmployeeList

```
D:\ANG-2025\Day10-EmpCrud\empMgtApp>ng g c components/EmployeeList
CREATE src/app/components/employee-list/employee-list.spec.ts (594 bytes)
CREATE src/app/components/employee-list/employee-list.ts (223 bytes)
CREATE src/app/components/employee-list/employee-list.css (0 bytes)
CREATE src/app/components/employee-list/employee-list.html (29 bytes)
```

>ng g c components/AddEmployee

```
D:\ANG-2025\Day10-EmpCrud\empMgtApp>ng g c components/AddEmployee
CREATE src/app/components/add-employee/add-employee.spec.ts (587 bytes)
CREATE src/app/components/add-employee/add-employee.ts (219 bytes)
CREATE src/app/components/add-employee/add-employee.css (0 bytes)
CREATE src/app/components/add-employee/add-employee.html (28 bytes)
```

>ng g c components/EditEmployee

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```
D:\ANG-2025\Day10-EmpCrud\empMgmtApp>ng g c components/EditEmployee
CREATE src/app/components/edit-employee/edit-employee.spec.ts (594 bytes)
CREATE src/app/components/edit-employee/edit-employee.ts (223 bytes)
CREATE src/app/components/edit-employee/edit-employee.css (0 bytes)
CREATE src/app/components/edit-employee/edit-employee.html (29 bytes)
```

>ng g c components/EmployeeDetails

```
D:\ANG-2025\Day10-EmpCrud\empMgmtApp>ng g c components/EmployeeDetails
CREATE src/app/components/employee-details/employee-details.spec.ts (615 bytes)
CREATE src/app/components/employee-details/employee-details.ts (235 bytes)
CREATE src/app/components/employee-details/employee-details.css (0 bytes)
CREATE src/app/components/employee-details/employee-details.html (32 bytes)
```

First setup routing in application. Go to app.routes.ts and modify as:

```
import { Routes } from '@angular/router';
import { EmployeeList } from '../components/employee-list/employee-list';
import { AddEmployee } from '../components/add-employee/add-employee';
import { EditEmployee } from '../components/edit-employee/edit-employee';
import { EmployeeDetails } from '../components/employee-details/employee-details';

export const routes: Routes = [
  { path: '', redirectTo: 'employees', pathMatch: 'full' },
  { path: 'employees', component: EmployeeList },
  { path: 'employees/add', component: AddEmployee },
  { path: 'employees/edit/:id', component: EditEmployee },
  { path: 'employees/:id', component: EmployeeDetails },
  { path: '**', redirectTo: 'employees' }
];
```

Check app.config.ts

```
import { ApplicationConfig, provideBrowserGlobalErrorListeners,
provideZoneChangeDetection } from '@angular/core';
import { provideRouter } from '@angular/router';

import { routes } from './app.routes';
import { provideHttpClient } from '@angular/common/http';

export const appConfig: ApplicationConfig = {
  providers: [
```



```
provideBrowserGlobalErrorListeners(),
provideZoneChangeDetection({ eventCoalescing: true }),
provideRouter(routes),
provideHttpClient()
]
};
```

Let's create a layout component to provide navbar

```
D:\ANG-2025\Day10-EmpCrud\empMgmtApp\src\app\Layout>ng g c Layout
CREATE src/app/Layout/layout/layout.spec.ts (551 bytes)
CREATE src/app/Layout/layout/layout.ts (196 bytes)
CREATE src/app/Layout/layout/layout.css (0 bytes)
CREATE src/app/Layout/layout/layout.html (22 bytes)
```

In Layout.ts

```
import { Component } from '@angular/core';
import { RouterLink, RouterOutlet } from '@angular/router';

@Component({
  selector: 'app-layout',
  imports: [RouterOutlet, RouterLink],
  templateUrl: './layout.html',
  styleUrls: ['./layout.css']
})
export class Layout {
}
```

In Layout.html

```
<nav class="navbar navbar-expand-lg navbar-dark bg-dark">
  <div class="container-fluid">
    <a class="navbar-brand" routerLink="/">EmployeeApp</a>
    <button class="navbar-toggler" type="button" data-bs-
toggle="collapse" data-bs-target="#navbarNav">
      <span class="navbar-toggler-icon"></span>
    </button>
    <div class="collapse navbar-collapse" id="navbarNav">
      <ul class="navbar-nav">
        <li class="nav-item">
          <a class="nav-link" routerLink="/employees"
routerLinkActive="active">Employees</a>
        </li>
```

```
<li class="nav-item">
  <a class="nav-link" routerLink="/employees/add"
routerLinkActive="active">Add Employee</a>
</li>
</ul>
</div>
</div>
</nav>
<div class="container mt-4">
  <router-outlet></router-outlet>
</div>
```

App.ts

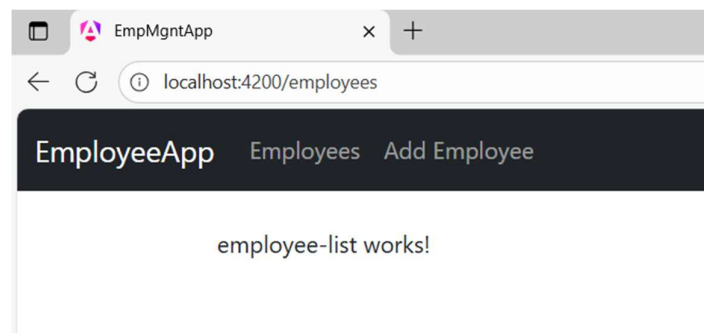
```
import { Component, signal } from '@angular/core';
import { RouterOutlet } from '@angular/router';
import { Layout } from "../Layout/layout/layout";

@Component({
  selector: 'app-root',
  imports: [RouterOutlet, Layout],
  templateUrl: './app.html',
  styleUrls: ['./app.css']
})
export class App {
  protected readonly title = signal('empMgntApp');
}
```

App.html

```
<app-layout></app-layout>
<router-outlet />
```

Run and Test once



Employee List Component

employeeelist.ts

```
import { CommonModule } from '@angular/common';
import { Component, OnInit } from '@angular/core';
import { RouterLink } from '@angular/router';
import { Employee } from '../employee';
import { EmployeeService } from '../services/employee-service';

@Component({
  selector: 'app-employee-list',
  imports: [CommonModule, RouterLink],
  templateUrl: './employee-list.html',
  styleUrls: ['./employee-list.css']
})
export class EmployeeList implements OnInit {
  employees: Employee[] = [];

  constructor(private empService: EmployeeService) { }

  ngOnInit() {
    this.loadEmployees();
  }

  loadEmployees() {
    this.empService.getEmployees().subscribe(data => this.employees
= data);
  }

  deleteEmployee(id: number) {
    this.empService.deleteEmployee(id).subscribe(() =>
this.loadEmployees());
  }
}
```

employee-list.html

```
<h2 class="mb-3">Employee List</h2>
<table class="table table-bordered table-hover">
  <thead class="table-dark">
    <tr>
      <th>ID</th>
```



```
<th>Name</th>
<th>Role</th>
<th>Salary</th>
<th>Actions</th>
</tr>
</thead>
<tbody>
  @for (emp of employees; track emp.id) {
    <tr>
      <td>{{ emp.id }}</td>
      <td>{{ emp.name }}</td>
      <td>{{ emp.role }}</td>
      <td>₹{{ emp.salary }}</td>
      <td>
        <a [routerLink]="['/employees', emp.id]" class="btn btn-sm
btn-info me-1">View</a>
        <a [routerLink]="['/employees/edit', emp.id]" class="btn
btn-sm btn-warning me-1">Edit</a>
        <button (click)="deleteEmployee(emp.id!)" class="btn btn-sm
btn-danger">Delete</button>
      </td>
    </tr>
  }
</tbody>
</table>
```

Add New Employee

add-employee.ts

```
import { CommonModule } from '@angular/common';
import { Component, OnInit } from '@angular/core';
import { FormBuilder, FormGroup, ReactiveFormsModule, Validators }
from '@angular/forms';
import { EmployeeService } from '../../services/employee-service';
import { Router } from '@angular/router';

@Component({
  selector: 'app-add-employee',
  imports: [ReactiveFormsModule, CommonModule],
  templateUrl: './add-employee.html',
```

```
styleUrl: './add-employee.css'
}))

export class AddEmployee implements OnInit {
  empForm!: FormGroup;

  constructor(private fb: FormBuilder, private empService:
EmployeeService, private router: Router) { }

  ngOnInit(): void {
    this.empForm = this.fb.group({
      name: ['', Validators.required],
      role: ['', Validators.required],
      salary: ['', [Validators.required, Validators.min(1000)]],
    });
  }

  onSubmit() {
    if (this.empForm.valid) {
      this.empService.addEmployee(this.empForm.value).subscribe(()
=> {
        this.router.navigate(['/employees']);
      });
    }
  }
}
```

add-employee.html

```
<h2>Add Employee</h2>
<form [formGroup]="empForm" (ngSubmit)="onSubmit()">
  <input type="text" placeholder="Name" formControlName="name" />
  <input type="text" placeholder="Role" formControlName="role" />
  <input type="number" placeholder="Salary" formControlName="salary"
/>
  <button type="submit">Add</button>
</form>
```

edit-employee.ts

```
import { Component, OnInit } from '@angular/core';
```

```
import { ActivatedRoute, Router } from '@angular/router';
import { EmployeeService } from '../services/employee-service';
import { Employee } from '../employee';
import { FormBuilder, FormGroup, ReactiveFormsModule, Validators }
from '@angular/forms';
import { CommonModule } from '@angular/common';
@Component({
  selector: 'app-edit-employee',
  imports: [ReactiveFormsModule, CommonModule],
  templateUrl: './edit-employee.html',
  styleUrls: ['./edit-employee.css']
})
export class EditEmployee implements OnInit {
  empForm!: FormGroup;
  id!: any;
  constructor(
    private route: ActivatedRoute,
    private router: Router,
    private empService: EmployeeService,
    private fb: FormBuilder
  ) { }
  ngOnInit() {
    this.empForm = this.fb.group({
      name: ['', Validators.required],
      role: ['', Validators.required],
      salary: ['', [Validators.required, Validators.min(1000)]],
    });

    this.id = this.route.snapshot.paramMap.get('id');
    this.empService.getEmployeeById(this.id).subscribe(emp => {
      this.empForm.patchValue(emp);
    });
  }

  onSubmit() {
    if (this.empForm.valid) {
      this.empService.updateEmployee(this.id, this.empForm.value as
Employee).subscribe(() => {
        this.router.navigate(['/employees']);
      });
    }
  }
}
```

```
}  
  
}
```

edit-employee.html

```
<h2 class="mb-3">Edit Employee</h2>  
<form [formGroup]="empForm" (ngSubmit)="onSubmit()" class="row g-3">  
  <div class="col-md-4">  
    <label class="form-label">Name</label>  
    <input type="text" class="form-control" formControlName="name">  
  </div>  
  <div class="col-md-4">  
    <label class="form-label">Role</label>  
    <input type="text" class="form-control" formControlName="role">  
  </div>  
  <div class="col-md-4">  
    <label class="form-label">Salary</label>  
    <input type="number" class="form-control"  
formControlName="salary">  
  </div>  
  <div class="col-12">  
    <button type="submit" class="btn btn-primary">Update  
Employee</button>  
  </div>  
</form>
```

employee-details.ts

```
import { CommonModule } from '@angular/common';  
import { Component, OnInit } from '@angular/core';  
import { Employee } from '../employee';  
import { ActivatedRoute, RouterLink } from '@angular/router';  
import { EmployeeService } from '../services/employee-service';  
  
@Component({  
  selector: 'app-employee-details',  
  imports: [CommonModule, RouterLink],  
  templateUrl: './employee-details.html',  
  styleUrls: ['./employee-details.css']  
})  
export class EmployeeDetails implements OnInit {  
  employee?: Employee;
```

```

    constructor(private route: ActivatedRoute, private empService:
EmployeeService) { }

    ngOnInit() {
        const id = this.route.snapshot.paramMap.get('id');
        this.empService.getEmployeeById(id).subscribe(emp =>
this.employee = emp);
    }
}

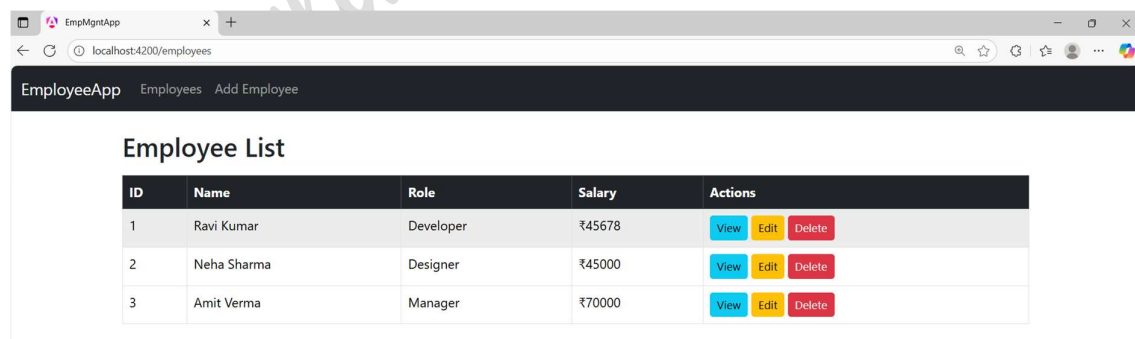
```

employee-details.html

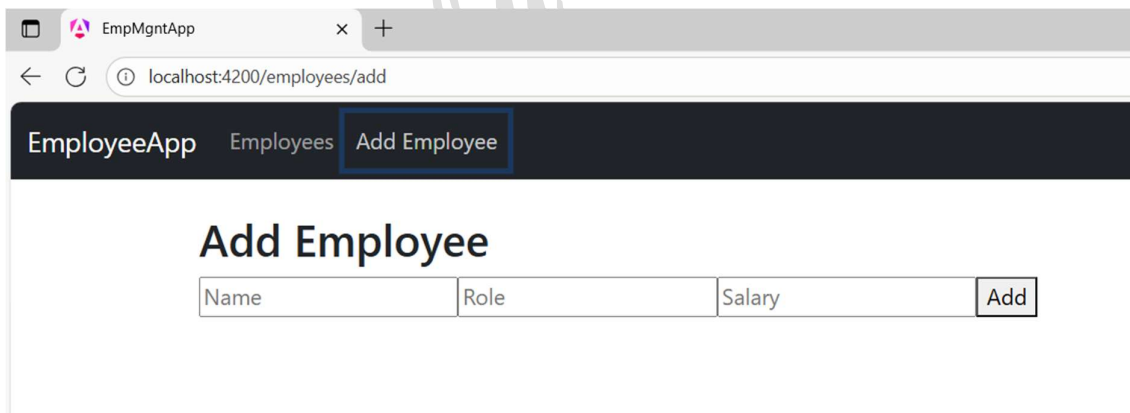
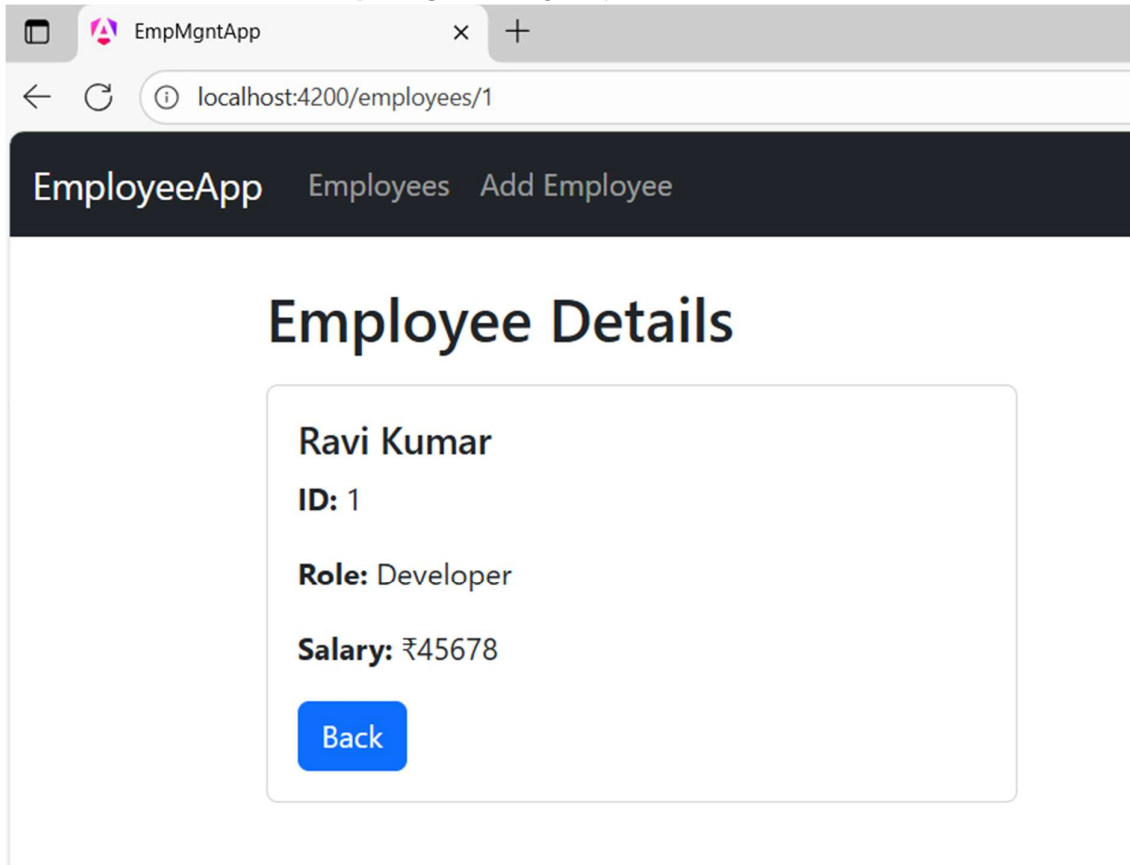
```

<h2 class="mb-3">Employee Details</h2>
@if(employee){
<div class="card" style="max-width: 400px">
    <div class="card-body">
        <h5 class="card-title">{{ employee.name }}</h5>
        <p class="card-text"><b>ID:</b> {{ employee.id }}</p>
        <p class="card-text"><b>Role:</b> {{ employee.role }}</p>
        <p class="card-text"><b>Salary:</b> ₹{{ employee.salary }}</p>
        <a routerLink="/employees" class="btn btn-primary">Back</a>
    </div>
</div>
}

```



ID	Name	Role	Salary	Actions
1	Ravi Kumar	Developer	₹45678	View Edit Delete
2	Neha Sharma	Designer	₹45000	View Edit Delete
3	Amit Verma	Manager	₹70000	View Edit Delete



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